

1 No-Cloning Theorem

1.1 Motivation

Here we are going to talk about one of the most fundamental theorems in quantum computing. But rather than just diving straight into the definition, lets talk about the need for it.

I am going to use some fundamental concepts of quantum computing to explain stuff. So if there is something that you are not familiar with then I strongly recommend reading about it from the internet (this will enhance your understanding of the field in general).

1.2 Understanding

Lets consider a qubit in $|0\rangle$ and then we apply a Hadamard gate to this qubit. This will transform the state to $|+\rangle$.

The circuit will look like this:

$$|0\rangle \text{ --- } \boxed{H} \text{ --- } |+\rangle$$

So what does a quantum state of $|+\rangle$ even mean?

We also know that

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

In other words, if we measure this state then there is a 50% chance of getting $|0\rangle$ and a 50% chance of getting $|1\rangle$.

This uncovers some fascinating lines of thought. So, like the above example, lets consider a random quantum state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

So we can just run a Monte-Carlo simulation (don't worry if you think this is the name of a brand).

It basically means that we will use the fundamental way of calculating probability. Like tossing a coin, if we do multiple tosses then the probability will tend to its theoretical value, which is 0.5.

Similarly, if we had access to a large number of identical copies of this quantum state then by measuring those copies multiple times we could estimate the probabilities associated with α and β . By performing measurements in different bases we could even reconstruct the complete quantum state.

Think about this. It almost feels like we are breaking some fundamental law here. Because if this line of thought is correct then there is no such thing as an unknown quantum state.

The fundamental law we are breaking here is the No-Cloning Theorem.

To do these calculations we require a large number of copies of the same unknown quantum state because on measuring the quantum state it will collapse into either $|0\rangle$ or $|1\rangle$, breaking the superposition.

And getting this large number of identical copies of an unknown quantum state is not possible because of the No-Cloning Theorem.

Now lets formally define it.

1.3 Definition

The No-Cloning Theorem states that:

”It is impossible to create an exact copy of an arbitrary unknown quantum state.”

In simpler words, there does not exist a universal quantum machine which can take an unknown quantum state and produce two identical copies of it.

If such a machine existed, then for every quantum state $|\psi\rangle$ it would perform the transformation

$$|\psi\rangle |0\rangle \longrightarrow |\psi\rangle |\psi\rangle .$$

The No-Cloning Theorem tells us that such a universal transformation is impossible.

1.4 Why Does This Matter?

At first glance this might seem like a strange restriction because copying classical information is something we do all the time.

However, this theorem is one of the reasons quantum information behaves so differently from classical information.

Some important consequences of this theorem are:

- An unknown quantum state cannot be copied perfectly.
- The idea we discussed earlier of creating many copies of a quantum state and then measuring them is not possible.
- Quantum messages cannot be copied in the same way that we copy files, photos, or classical bits.
- This property plays a very important role in quantum cryptography and helps make quantum communication secure.

For now, this level of understanding is more than enough. The main takeaway from this chapter is simple: an arbitrary unknown quantum state cannot be copied perfectly.

If you’re curious about the mathematical proof, feel free to check out the QuEra article on the No-Cloning Theorem or Nielsen and Chuang’s *Quantum Computation and Quantum Information*.